

Team Note of Baka Team

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1 Template

1.1 Template

```

#include <bits/stdc++.h>

using namespace std;
using ll = long long;

#define MASK(k) 1LL << (k)
#define BIT(x, k) ((x) >> (k) & 1)
#define all(x) (x).begin(), (x).end()

template<class T> bool minimize(T& a, const T& b) {
    if(a > b) return a = b, true;
    return false;
}

template<class T> bool maximize(T& a, const T& b) {
    if(a < b) return a = b, true;
    return false;
}

ll fdiv(ll a, ll b) {
    assert(b != 0);
    if(b < 0) a *= -1, b *= -1;
    return a >= 0 ? a / b : (a + 1) / b - 1;
}

ll cdiv(ll a, ll b) {
    assert(b != 0);
    if(b < 0) a *= -1, b *= -1;
    return a <= 0 ? a / b : (a - 1) / b + 1;
}

```

1.2 Debug

```

template<class Tp1, class Tp2>
ostream& operator << (ostream& cout, pair<Tp1, Tp2> val) {
    return cout << "(" << val.first << " " << val.second <<
        ")";
}

template<class Data, class Tp =
decltype(declval<Data>().begin())>
typename enable_if<!is_same<Data, string>::value,
    ostream&>::type
operator << (ostream& cout, Data val) {
    cout << "[";
    for(auto i = val.begin(); i != val.end(); ++i)
        cout << (i == val.begin() ? "" : " ") << *i;
    return cout << "]";
}

```

```

template<class Data, class = decltype(declval<Data>().top())>
ostream& operator << (ostream& cout, Data val) {
    cout << "[";
    for(; val.size(); val.pop())
        cout << val.top() << (val.size() == 1 ? "" : " ");
    return cout << "]";
}

template<class Tp> ostream& operator << (ostream& cout,
queue<Tp> val) {
    cout << "[";
    for(; val.size(); val.pop())
        cout << val.front() << (val.size() == 1 ? "" : " ");
    return cout << "]";
}

void debug_utils() {}

template<class T, class ...U> void debug_utils(T a, U... b) {
    cerr << a;
    if(sizeof...(b)) { cerr << ", "; debug_utils(b...);}
}

#define debug(...) { cerr << #__VA_ARGS__ << " = ";
debug_utils(__VA_ARGS__); cerr << "\n"; }

void print_utils(string& result, int ptr, const string& str) {
    for(; ptr < (int) str.size(); ++ptr)
        result.push_back(str[ptr]);
}

template<class T, class ...U> void print_utils(string& result,
int ptr, const string& str, T a, U... b) {
    for(; ptr < (int) str.size(); ++ptr) {
        if(str[ptr] == '{' && str[ptr + 1] == '}') break;
        result.push_back(str[ptr]);
    }
    stringstream stream;
    stream << a;
    result += stream.str();
    if(sizeof...(b)) { cerr << ", "; print_utils(result, ptr +
2, str, b...); }
    else { print_utils(result, ptr + 2, str);
}
}

#define print(...) { string result = ""; print_utils(result,
0, __VA_ARGS__); cerr << result << "\n"; }

```

1.3 Generate

```

mt19937_64
rng(chrono::steady_clock::now().time_since_epoch().count());

ll randint(ll a, ll b) {
    return uniform_int_distribution<ll>(a, b)(rng);
}

```

1.4 Stress Test (Linux)

```

memory=524288 # 512MB
stack_size=524288 # 512MB

ulimit -v "${memory}"
ulimit -s "${stack_size}"

```

```

g++ main.cpp -DSTRESS -O2 -std=c++17 -Wno-unused-result
-Wshadow -Wall -o main
g++ brute.cpp -DSTRESS -O2 -std=c++17 -Wno-unused-result
-Wshadow -Wall -o brute
g++ gen.cpp -DSTRESS -O2 -std=c++17 -Wno-unused-result -o gen

```

```

for((i = 1; i <= 1000; ++i)); do {
    ./gen > test.in
    ./main < test.in > test.out
}

```

```

./brute < test.in > test.ans

# External grading program:
# checker > /dev/null || {
# General:
diff -Z test.out test.ans > /dev/null || {
    echo "WA on the following test:"
    cat test.in
    echo "=====
    echo "Your answer is:"
    cat test.out
    echo "=====
    echo "Correct answer is:"
    cat test.ans
    break
}
echo "Passed test: " $i
}
done

```

```

rm main
rm gen
rm brute

```

1.5 Stress Test (Window)

```

#include <bits/stdc++.h>
using namespace std;

void showFile(const char* name) {
    ifstream file(name);
    cout << file.rdbuf() << endl;
}

int stressTest() {
    if (system("g++ main.cpp -DSTRESS -O2 -std=c++17 -o
stress_main.exe") != 0 ||
        system("g++ brute.cpp -DSTRESS -O2 -std=c++17 -o
stress_brute.exe") != 0 ||
        system("g++ gen.cpp -DSTRESS -O2 -std=c++17 -o
stress_gen.exe") != 0) {
        cerr << "Compilation failed." << endl;
        return 1;
    }

    for (int i = 1; i <= 1000; ++i) {
        if (system(".\\stress_gen.exe > test.in") != 0 ||
            system(".\\stress_main.exe < test.in > test.out")
            != 0 ||
            system(".\\stress_brute.exe < test.in > test.ans")
            != 0) {
            cerr << "Program failed on test " << i << "." <<
endl;
            return 1;
        }

        if (system("fc /B test.out test.ans > nul") != 0) {
            cout << "WA on test " << i << ":" << endl;
            cout << "Input:" << endl;
            showFile("test.in");
            cout << "Your answer:" << endl;
            showFile("test.out");
            cout << "Correct answer:" << endl;
            showFile("test.ans");
            return 1;
        }

        cout << "Passed test: " << i << endl;
    }
    return 0;
}

int main() {
    int result = stressTest();
    remove("stress_main.exe");
    remove("stress_brute.exe");
    remove("stress_gen.exe");
    return result;
}

```

1.6 Pragma

```
#pragma GCC optimize("unroll-loops,02,fast-math,
no-stack-protector")
#pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
```

2 Data Structure

2.1 Mutidimensional Vector

```
template<class Tp, int D = 1>
struct Tvector : public vector<Tvector<Tp, D - 1>> {
    template <class... Args>
    Tvector(int n = 0, Args... args) : vector<Tvector<Tp, D - 1>>(n, Tvector<Tp, D - 1>(args...)) {}
};

template <class Tp>
struct Tvector<Tp, 1> : public vector<Tp> {
    Tvector(int n = 0, Tp val = Tp()) : vector<Tp>(n, val) {}
};
```

2.2 Rollback

```
vector<pair<int*, int>> event;

void assign(int* u, int v) {
    event.push_back({u, exchange(*u, v)});
}

void rollback(int t) {
    for(; (int) event.size() > t; event.pop_back()) {
        *event.back().first = event.back().second;
    }
}
```

2.3 Compress

```
template<class Tp> struct Compress : vector<Tp> {
    Compress() = default;
    template<class Data, class =
    decltype(*declval<Data>().begin())>
    Compress(const Data& data) : Compress(data.begin(),
    data.end()) {}
    template<class Iterator, class =
    decltype(*declval<Iterator>())>
    Compress(Iterator first, Iterator last) {
        for(auto it = first; it != last; ++it)
            (*this).push_back(*it);
        build();
    }
    void build() {
        sort(begin(*this), end(*this));
        (*this).erase(unique(begin(*this), end(*this)),
        end(*this));
    }
    int prod(const Tp& x, int default_value = -1) {
        auto it = lower_bound(begin(*this), end(*this), x);
        return it != end(*this) && *it == x ? it -
        begin(*this) : default_value;
    }
};
```

2.4 BIT

```
int n;
vector<int> bit1, bit2;
/*
    updateRange(l, r, v): Moi phan tu cong them v
    rangeSum(l, r): Tong cac phan tu tu l den r
*/

void init(int n_init) {
    n = n_init;
    bit1.assign(n+1, 0);
    bit2.assign(n+1, 0);
}
```

```
}

void updatePoint(vector<int> &b, int u, int v) {
    int idx = u;
    while (idx <= n) {
        b[idx] += v;
        idx += (idx & (-idx));
    }
}

void updateRange(int l, int r, int v) {
    updatePoint(bit1, l, (n - l + 1) * v);
    updatePoint(bit1, r + 1, -(n - r) * v);
    updatePoint(bit2, l, v);
    updatePoint(bit2, r + 1, -v);
}

int getSumOnBIT(vector<int> &b, int u) {
    int idx = u, ans = 0;
    while (idx > 0) {
        ans += b[idx];
        idx -= (idx & (-idx));
    }
    return ans;
}

int prefixSum(int u) {
    return getSumOnBIT(bit1, u) - getSumOnBIT(bit2, u) * (n -
    u);
}

int rangeSum(int l, int r) {
    return prefixSum(r) - prefixSum(l - 1);
}

// first index a[0] + ... + a[index] >= s.
int lowerBoundPrefix(long long s) {
    int idx = 0;
    long long sum1 = 0, sum2 = 0;

    int k = 1;
    while ((k << 1) <= n) k <<= 1;

    for (; k > 0; k >>= 1) {
        int nxt = idx + k;

        if (nxt <= n) {
            long long cur1 = sum1 + bit1[nxt];
            long long cur2 = sum2 + bit2[nxt];
            long long pref = cur1 - cur2 * (n - nxt);
            if (pref < s) { idx = nxt; sum1 = cur1; sum2 =
            cur2; }
        }

        int ans = idx + 1;
        return (ans <= n) ? ans : -1;
    }
}
```

2.5 BIT 2D

```
template<class Tp, class Index> struct FenwickTree2D :
vector<pair<Index, Index>> {
    void build() {
        for(const auto& [u, v] : (*this)) {
            cx.push_back(u); cy.push_back(v);
        } cx.build(); cy.build();
        nodes.assign(int(cx.size()), vector<int>{});
        bit.assign(int(cx.size()), vector<int>{});
        for(const auto& [u, v] : (*this)) {
            fake_update(u, v);
        }
        for(int i = 0; i < int(cx.size()); ++i) {
            nodes[i].build();
            bit[i].assign(int(nodes[i].size()), 0);
        }
    }

    void add(int u, int v, Tp val) {
        assert(~u && ~v);
    }
};
```

```

    for(int x = u + 1; x <= int(cx.size()); x += (x & -x))

        for(int y = lower_bound(begin(nodes[x - 1]),
            end(nodes[x - 1]), v) - begin(nodes[x - 1]) + 1; y
            <= int(nodes[x - 1].size()); y += (y & -y))
            bit[x - 1][y - 1] += val;
    }

    void cadd(Index u, Index v, Tp val) {
        add(cx.prod(u), cy.prod(v), val);
    }

    Tp csum(Index u, Index v) {
        return sum(cx.prod(u), cy.prod(v));
    }

    Tp csum(Index x, Index y, Index u, Index v) {
        return sum(cx.prod(x), cy.prod(y), cx.prod(u),
            cy.prod(v));
    }

    Tp sum(int u, int v) {
        Tp ret {};
        for(int x = u + 1; x > 0; x -= (x & -x))
            for(int y = upper_bound(begin(nodes[x - 1]),
                end(nodes[x - 1]), v) - begin(nodes[x - 1]); y >
                0; y -= (y & -y))
                ret += bit[x - 1][y - 1];
        print("%d %d = %d", u, v, ret);
        return ret;
    }

    Tp sum(int x, int y, int u, int v) {
        return sum(u, v) + sum(x-1, y-1) - sum(u, y-1) -
            sum(x-1, v);
    }

    void fake_update(Index u, Index v) {
        u = cx.prod(u); v = cy.prod(v);
        for(++u; u <= int(cx.size()); u += (u & -u))
            nodes[u - 1].push_back(v);
    }

    vector<vector<Tp>> bit;
    Compress<Tp> cx, cy;
    vector<Compress<Tp>> nodes;
};

// Usage:
// push_back({x, y})
// add(x, y) cadd(x, y)
// sum(u, v) csum(u, v)
// sum(x, y, u, v) csum(x, y, u, v)

```

2.6 Spare Table

```

const int N = 100100;
const int K = 22;

int n;
int a[N];
int rmq[N][K];

void build() {
    for(int i = 1; i <= n; ++i) {
        rmq[i][0] = a[i];
    }
    for(int k = 1; (1 << k) <= n; ++k)
        for(int i = 1; i + (1 << (k - 1)) <= n; ++i) {
            rmq[i][k] = min(rmq[i][k - 1], rmq[i + (1 << (k - 1))][k - 1]);
        }
}

int get(int l, int r) {
    int pot = 31 - __builtin_clz(r - l + 1);
    return min(rmq[l][pot], rmq[r - (1 << pot) + 1][pot]);
}

```

2.7 Wavelet Tree

```

struct wavelet {
    wavelet *left, *right;
    vector<ll> pref;
    int wl, wr;

    wavelet() {}

    wavelet(int tl, int tr, int pL, int pR, vector<ll> &v) {
        wl = tl, wr = tr;
        if (wl == wr || pL > pR) return;

        int mid = (wl + wr) >> 1;

        pref.pb(0);
        for (int i = pL; i <= pR; i++)
            pref.pb(pref.back() + (v[i] <= mid));

        ll piv = stable_partition(v.begin() + pL, v.begin() +
            pR + 1
            , [&](int x){ return x <= mid; }) -
            v.begin() - 1;

        left = new wavelet(wl, mid, pL, piv, v);
        right = new wavelet(mid + 1, wr, piv + 1, pR, v);
    }

    ll findKth(int k, int l, int r) {
        if (wl == wr) return wl;
        // cout << wl << " " << wr << " " << k << '\n';

        int amt = pref[r] - pref[l - 1];
        int lBound = pref[l - 1];
        int rBound = pref[r];

        if (amt >= k) return left->findKth(k, lBound + 1,
            rBound);

        return right->findKth(k - amt, l - lBound, r -
            rBound);
    }
};

```

2.8 Sparse lichao tree

```

struct Line {
    ll m, b;
    Line(ll _m = 0, ll _b = INF * 8) : m(_m), b(_b) {}
};

ll F(Line l, ll x) {
    return l.m * x + l.b;
}

struct lichao_t {
    lichao_t *left = nullptr, *right = nullptr;
    Line mn;
    lichao_t(ll tl = 0, ll tr = 0) {}
    void Update(ll tl, ll tr, Line nLine) {
        ll mid = (tl + tr) >> 1;
        bool pLeft = (F(nLine, tl) < F(mn, tl));
        bool pMid = (F(nLine, mid) < F(mn, mid));
        if (pMid)
            swap(mn, nLine);
        if (tl == tr)
            return;
        if (pLeft != pMid) {
            if (left == nullptr)
                left = new lichao_t();
            left->Update(tl, mid, nLine);
        } else {
            if (right == nullptr)
                right = new lichao_t();
            right->Update(mid + 1, tr, nLine);
        }
    }

    ll Query(ll tl, ll tr, ll x) {
        if (tl == tr)
            return F(mn, x);
        ll mid = (tl + tr) >> 1;
        ll res = F(mn, x);
        if (x <= mid) {

```

```

    if (left != nullptr)
        minimize(res, left->Query(tl, mid, x));
} else {
    if (right != nullptr)
        minimize(res, right->Query(mid + 1, tr, x));
}
return res;
}
};

```

2.9 Line Container

```

struct Line {
    mutable ll k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
    bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
    // (for doubles, use inf = 1/.0, div(a,b) = a/b)
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b) { // floored division
        return a / b - ((a ^ b) < 0 && a % b); }
    bool isect(iterator x, iterator y) {
        if (y == end()) return x->p = inf, 0;
        if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
        else x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }
    void add(ll k, ll m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p)
            isect(x, erase(y));
    }
    ll query(ll x) {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};

```

2.10 Heavy Light Decomposition

```

int dep[N], parent[N], sz[N];
int head[N], tin[N], tout[N], idx[N], cnt = 0;

void dfs(int u) {
    sz[u] = 1;
    for(int& v : G[u]) {
        parent[v] = u;
        dep[v] = dep[u] + 1;
        G[v].erase(find(all(G[v]), u));
        dfs(v);
        sz[u] += sz[v];
        if(sz[G[u][0]] < sz[v]) swap(G[u][0], v);
    }
}

void hld(int u, int h) {
    idx[tin[u] = ++cnt] = u;
    head[u] = h;
    for(const int& v : G[u])
        hld(v, G[u][0] == v ? h : v);
    tout[u] = cnt;
}

void hld_update(int u, int v, int w) {
    for(; head[u] != head[v]; v = parent[head[v]]) {
        if(dep[head[u]] > dep[head[v]]) swap(u, v);
        // update(tin[head[v]], tin[v], w);
    }
    if(dep[u] > dep[v]) swap(u, v);
    // update(tin[u], tin[v], w)
}

int hld_prod(int u, int v) {
    int ret = ?;

```

```

for(; head[u] != head[v]; v = parent[head[v]]) {
    if(dep[head[u]] > dep[head[v]]) swap(u, v);
    // ret = op(ret, prod(tin[head[v]], tin[v]));
}
if(dep[u] > dep[v]) swap(u, v);
// ret = op(ret, prod(tin[u], tin[v]))
return ret;
}

```

2.11 Ordered Set

```

#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>

using namespace __gnu_pbds;

template<class T>
using OrderedTree = tree<T, null_type, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;
template<class T>
using OrderedMultiTree = tree<T, null_type, less_equal<T>,
    rb_tree_tag,
    tree_order_statistics_node_update>;

// [void] s.insert(x): insert x into s
// [itr ] s.find_by_order(id): s[id]
// [int ] s.order_of_key(x): min_id; s[min_id] >= x or number
// val < x
// [int ] s.order_of_key(x): min_id; s[min_id] >= x or number
// val < x
// [int ] s.order_of_key(r + 1) - s.order_of_key(l): number
// val in range [l, r]
// [void] s.erase(x): erase x in s
// [itr ] s.find(x): find x in s. if not exist, return s.end()
// [itr ] s.lower_bound(x): s[min_id]; s[min_id] >= x
// [itr ] s.upper_bound(x): s[min_id]; s[min_id] > x

```

2.12 Merge Sort Tree

```

template<class Tp>
struct MergeSortTree : vector<Tp> {
    void build() {
        n = this->size();
        st.resize((n<<2) + 1, {});
        build(1, 0, n-1);
    }
    int query(int s, int t, int from, int to) {
        return query(1, 0, n-1, s, t, from, to);
    }
    void build(int id, int l, int r) {
        if(l == r) { st[id].push_back((*this)[l]); return; }
        int mid = (l+r) >> 1;
        build(id<<1, l, mid);
        build(id<<1|1, mid+1, r);
        st[id].resize(r-l+1);
        merge(st[id<<1].begin(), st[id<<1].end(),
            st[id<<1|1].begin(), st[id<<1|1].end(),
            st[id].begin());
    }

    int query(int id, int l, int r, int s, int t, int from,
        int to) {
        if(l > t || r < s) return 0;
        if(s <= l && r <= t)
            return upper_bound(st[id].begin(), st[id].end(), to)
                - lower_bound(st[id].begin(), st[id].end(), from);
        int mid = (l+r) >> 1;
        return query(id<<1, l, mid, s, t, from, to) +
            query(id<<1|1, mid+1, r, s, t, from, to);
    }

    int n;
    vector<vector<Tp>> st;
};

```

2.13 Segment Tree 2D

```
/**

```

```

* Usage:
* a[0..n-1][0..n-1]
* build()
* update(x, y, val): a[x][y] += val
* query(x1, y1, x2, y2): a[x][y] x1 <= x <= x2, y1 <= y <= y2
*/
const int MAX=1003;

int a[MAX][MAX], seg[2*MAX][2*MAX], n;

void build() {
    for(int x=0;x<n;++x)
        for(int y=0;y<n;++y)
            seg[x+n][y+n] = a[x][y];
    for (int x = 2*n; x; x--) for (int y = 2*n; y; y--) {
        if (x < n) seg[x][y] = seg[2*x][y] + seg[2*x+1][y];
        if (y < n) seg[x][y] = seg[x][2*y] + seg[x][2*y+1];
    }
}

int query(int x1, int y1, int x2, int y2) {
    int ret = 0, y3 = y1 + n, y4 = y2 + n;
    for (x1 += n, x2 += n; x1 <= x2; ++x1 /= 2, --x2 /= 2) {
        for (y1 = y3, y2 = y4; y1 <= y2; ++y1 /= 2, --y2 /= 2) {
            if (x1%2 == 1 and y1%2 == 1) ret += seg[x1][y1];
            if (x1%2 == 1 and y2%2 == 0) ret += seg[x1][y2];
            if (x2%2 == 0 and y1%2 == 1) ret += seg[x2][y1];
            if (x2%2 == 0 and y2%2 == 0) ret += seg[x2][y2];
        }
    }
    return ret;
}

void update(int x, int y, int val) {
    int y2 = y + n;
    for (x += n; x; x /= 2, y = y2) {
        if (x >= n) seg[x][y] = val;
        else seg[x][y] = seg[2*x][y] + seg[2*x+1][y];

        while (y /= 2) seg[x][y] = seg[x][2*y] + seg[x][2*y+1];
    }
}

```

2.14 Mod Segment Tree

```

const int N = 200200;
const int inf = 1 << 30;
int n, m, a[N];

struct Node {
    ll sum;
    int min, max, lazy;
    Node(int u = 0): sum(u), min(u), max(u), lazy(inf) {}
    Node(ll sum, int min, int max, int lazy): sum(sum),
        min(min), max(max), lazy(lazy) {}
} IT[4 * N];

Node operator + (Node a, Node b) {
    return {
        a.sum + b.sum,
        min(a.min, b.min),
        max(a.max, b.max),
        inf
    };
}

#define m ((l + r) >> 1)
void Build(int id, int l, int r) {
    if(l == r) {
        IT[id] = a[l];
        return;
    }
    Build(id << 1, l, m);
    Build(id << 1 | 1, m + 1, r);
    IT[id] = IT[id << 1] + IT[id << 1 | 1];
}

void Self(int id, int l, int r, int v) {
    IT[id].max = IT[id].min = v;
    IT[id].lazy = v;

```

```

    IT[id].sum = (ll) v * (r - l + 1);
}

void Push(int id, int l, int r) {
    if(IT[id].lazy == inf || l == r) return;
    Self(id << 1, l, m, IT[id].lazy);
    Self(id << 1 | 1, m + 1, r, IT[id].lazy);
    IT[id].lazy = inf;
}

void UpdateMod(int id, int l, int r, int s, int t, int mod) {
    Push(id, l, r);
    if(l > t || r < s || IT[id].max < mod) return;
    if(s <= l && r <= t && IT[id].max == IT[id].min) {
        int v = IT[id].max % mod;
        Self(id, l, r, v);
        return;
    }
    UpdateMod(id << 1, l, m, s, t, mod);
    UpdateMod(id << 1 | 1, m + 1, r, s, t, mod);
    IT[id] = IT[id << 1] + IT[id << 1 | 1];
}

void UpdateVal(int id, int l, int r, int u, int v) {
    Push(id, l, r);
    if(l > u || r < u) return;
    if(l == r) {
        IT[id] = v;
        return;
    }
    UpdateVal(id << 1, l, m, u, v);
    UpdateVal(id << 1 | 1, m + 1, r, u, v);
    IT[id] = IT[id << 1] + IT[id << 1 | 1];
}

ll GetSum(int id, int l, int r, int s, int t) {
    Push(id, l, r);
    if(l > t || r < s) return 0;
    if(s <= l && r <= t) return IT[id].sum;
    return GetSum(id << 1, l, m, s, t) + GetSum(id << 1 | 1, m
        + 1, r, s, t);
}
#undef m

```

2.15 Treap

```

struct Lazy {
    // [...]
};

struct Node {
    // [...]
};

Node operator + (Node u, Node v) {
    // [...]
    return Node{};
}

struct Tnode {
    Tnode *l = NULL, *r = NULL;
    Node node, key;
    Lazy lazy;
    int size = 1, prior = 0;
    Tnode(Node key = Node{}, int prior = randint(-1 << 30, 1
        << 30)): node(key), key(key), prior(prior) {}
};

using Pnode = Tnode*;

Pnode IgnoreNode;

// PreProcess.
void InitIgnoreNode() {
    IgnoreNode = new Tnode{};
    IgnoreNode->size = 0;
}

#define NODE(x) (x ? x : IgnoreNode)

Pnode FIX(Pnode u) {

```

```

    if(u) {
        u->size = NODE(u->l)->size + 1 + NODE(u->r)->size;
        u->node = NODE(u->l)->node + NODE(u->r)->key +
            NODE(u->r)->node;
    }
    return u;
}

void update_node(Pnode u, Lazy val) {
    if(!u) return;
    // [...]
}

void Down(Pnode t) {
}

Pnode merge(Pnode l, Pnode r) {
    if(!l || !r) return (l ? l : r);
    Down(l); Down(r);
    if(l->prior > r->prior) {
        l->r = merge(l->r, r);
        return FIX(l);
    } else {
        r->l = merge(l, r->l);
        return FIX(r);
    }
}

pair<Pnode, Pnode> split(Pnode t, int k) {
    if(!t) return {NULL, NULL};
    else Down(t);
    Pnode l = NULL, r = NULL;
    if(k <= NODE(t->l)->size) tie(l, t->l) = split(t->l, k), r = t;
    else tie(t->r, r) = split(t->r, k - 1 -
        NODE(t->l)->size), l = t;
    FIX(t);
    return {l, r};
}

tuple<Pnode, Pnode, Pnode> split(Pnode t, int u, int v) {
    if(!t) return {NULL, NULL, NULL};
    Pnode l = NULL, m = NULL, r = NULL;
    tie(t, r) = split(t, v + 1);
    tie(l, m) = split(t, u);
    return {l, m, r};
}

void DFS(Pnode t) {
    if(!t) return;
    Down(t);
    DFS(t->l);
    // [...]
    DFS(t->r);
}

```

2.16 Mod Int

namespace md

```

{
    using T = ll; // T : int, long long
    T mod = 1e9 + 7; //
    -----
    Modulus Number.
    int m = -1;
    struct Modulus
    {
        T x = T();
        inline void fix() { while(x >= mod) x -= mod; }
        Modulus(ll v = ll()) { x = v; fix(); }

        friend Modulus Mul(Modulus c, Modulus d); friend
        Modulus Pow(Modulus c, ll exp); friend Modulus
        Inv(Modulus c);
        Modulus operator + (const Modulus& c) { return
        Modulus(x + c.x); }
        Modulus operator - (const Modulus& c) { return
        Modulus(x - c.x + mod); }
        Modulus operator * (const Modulus& c) { return
        Mul(*this, c.x); }
    }
}

```

```

    Modulus operator / (const Modulus& c) { return
    Mul(*this, Inv(c)); }
    bool operator == (const Modulus& c) { return x == c.x;
    }
    bool operator != (const Modulus& c) { return x != c.x;
    }

    friend istream& operator >> (istream& cin, Modulus &c)
    {
        cin >> c.x; c.x = c.x % mod + mod;
        return c.fix(), cin;
    }

    friend ostream& operator << (ostream& cout, const
    Modulus &c) {
        return cout << c.x;
    }
}

vector<Modulus> Fact, Invert;

// [T = int] O(1) | [T == ll] O(log(exp))
Modulus Mul(Modulus c, Modulus d) {
    ll exp = d.x;
    if(sizeof(T) == 4UL) return Modulus(c.x * exp % mod);
    Modulus res = 0;
    for(; exp > 0; exp >= 1, c = c + c)
        if(exp & 1) res = res + c;
    return res;
}

// [T = int] O(log(exp)) | [T = ll] O(log(exp) * log(mod))
Modulus Pow(Modulus c, ll exp) {
    Modulus res = 1;
    for(; exp > 0; exp >= 1, c = c * c)
        if(exp & 1) res = res * c;
    return res;
}

// [c.x <= m] <O(m), O(1)> | <O(1), O(log(min(mod, c.x)))>
Modulus Inv(Modulus c) {
    if(c.x <= m)
        return Mul(Invert[c.x], Fact[c.x - 1]);
    ll a = c.x, b = mod, ax = 1, bx = 0;
    while(b > 0) {
        ll q = a / b, r = a % b;
        a = b, b = r;
        r = ax - bx * q;
        ax = bx, bx = r;
    }
    if(ax < 0) ax += mod;
    return Modulus(ax);
}

// O(m)
void BuildMOD(int M) {
    m = M;
    Fact = Invert = vector<Modulus>(m + 1, 0);
    Fact[0] = Invert[0] = 1;
    for(int i = 1; i <= m; ++i)
        Fact[i] = Fact[i - 1] * i;
    Invert[m] = Pow(Fact[m], mod - 2);
    for(int i = m - 1; i >= 1; --i)
        Invert[i] = Invert[i + 1] * ll(i + 1);
}

// <O(m), O(1)>
Modulus nCk(ll n, ll k) {
    Modulus res = 1;
    while(n != 0 || k != 0) {
        ll rn = n % mod, rk = k % mod;
        n /= mod, k /= mod;
        if(rn < rk) return 0;
        res = res * Fact[rn] / (Fact[rk] * Fact[rn - rk]);
    }
    return res;
}
} // namespace md
using namespace md;

```

2.17 Color Set

```
#include "bits/stdc++.h"
using namespace std;

struct ColorUpdate {
    struct Seg {
        int l, r, c;
        Seg(int l, int r, int c): l(l), r(r), c(c) {}
        bool operator < (const Seg& other) const { return l < other.l; }
        friend ostream& operator << (ostream& cout, const Seg& val) {
            return cout << "[" << val.l << " " << val.r <<
                "]" << val.c << " ";
        }
    };
    set<Seg> segs;
    void cut(int x) {
        auto it = segs.lower_bound({x, 0, 0});

        if(it == segs.begin()) return;
        --it;

        if(it->r == x - 1) return;

        Seg s = *it;
        segs.erase(s);
        segs.insert(Seg(s.l, x - 1, s.c));
        segs.insert(Seg(x, s.r, s.c));
    }

    void add(int l, int r, int c) {
        cut(l), cut(r + 1);
        Seg s(l, r, c);
        auto it = segs.lower_bound(s);
        while(it != segs.end() && it->l <= s.r) {
            auto it2 = it++;
            segs.erase(it2);
        }
        segs.insert(s);
    }

    void remove(int l, int r) {
        cut(l), cut(r + 1);
        Seg s(l, r, 0);
        auto it = segs.lower_bound(s);
        while(it != segs.end() && it->l <= s.r) {
            auto it2 = it++;
            segs.erase(it2);
        }
    }
};

// Usage:
// add(l, r, c)
// remove(l, r, c)
// cut(x)
```

3 Graphs

3.1 Bridge và Articulation Point

```
int n, m;
vector<int> adj[MAX_N];
int low[MAX_N], timeIn[MAX_N];
int dfsTime;
vector<pii> edge;
bool used[MAX_N];
int numCau, numKhop;

void dfs(int u) {
    timeIn[u] = low[u] = ++dfsTime;

    bool isKhop = false;
    int cnt = 0;

    for (int id : adj[u]) {
        if (used[id]) continue;
        used[id] = true;
```

```
int v = edge[id].fi ^ edge[id].se ^ u;
if (!timeIn[v]) {
    dfs(v);
    cnt++;
    minimize(low[u], low[v]);
    numCau += (low[v] > timeIn[u]);
    isKhop |= (low[v] >= timeIn[u]);
}
else {
    minimize(low[u], timeIn[v]);
}
}

if (timeIn[u] > 1) {
    numKhop += isKhop;
}
else {
    numKhop += (cnt > 1);
}
}

signed main() {
    cin >> n >> m;
    for (int i = 1; i <= m; i++) {
        int u, v;
        cin >> u >> v;
        adj[u].pb((int) edge.size());
        adj[v].pb((int) edge.size());
        edge.pb({u, v});
    }

    for (int i = 1; i <= n; i++) {
        if (!timeIn[i]) {
            dfsTime = 0;
            dfs(i);
        }
    }

    cout << numKhop << " " << numCau;

    return 0;
}
```

3.2 Strongly Connected Components

```
int n, m;
vector<int> adj[MAX_N];
int timeIn[MAX_N], low[MAX_N];
bool deleted[MAX_N];
int dfsTime = 0;
int numStronglyConnectedComponents = 0;
int root[MAX_N];
stack<int> st;

void dfs(int u) {
    timeIn[u] = low[u] = ++dfsTime;
    st.push(u);

    for (int v : adj[u]) {
        if (deleted[v]) continue;

        if (!timeIn[v]) {
            dfs(v);
            minimize(low[u], low[v]);
        }
        else {
            minimize(low[u], timeIn[v]);
        }
    }

    if (timeIn[u] == low[u]) {
        numStronglyConnectedComponents++;
        int v;
        do {
            v = st.top();
            st.pop();
            root[v] = u;
            deleted[v] = true;
        }
        while (v != u);
    }
}
```

```

}

signed main() {
    ios_base::sync_with_stdio(false);
    cin.tie(nullptr);

    cin >> n >> m;
    for (int i = 1; i <= m; i++) {
        int u, v;
        cin >> u >> v;
        adj[u].pb(v);
    }

    for (int i = 1; i <= n; i++) {
        if (!timeIn[i]) {
            dfs(i);
        }
    }

    cout << numStronglyConnectedComponents;

    return 0;
}

```

3.3 Lowest Common Ancestor

```

vector<int> G[N];
int ID[N << 1], tin[N], tout[N], counter = 0;

int dep[N], par[N][LOG];

void dfs(int u, int p = 0) {
    ID[ tin[u] = ++counter ] = u;
    dep[u] = dep[p] + 1;
    par[u][0] = p;
    for(int k = 1; k < LOG; ++k)
        par[u][k] = par[par[u][k - 1]][k - 1];
    for(const int& v : G[u])
        if(v != p) dfs(v, u);
    ID[ tout[u] = ++counter ] = u;
}

int lca(int u, int v) {
    if(dep[u] < dep[v]) swap(u, v);
    int diff = dep[u] - dep[v];
    for(int k = LOG - 1; k >= 0; --k)
        if(diff & MASK(k)) u = par[u][k];
    if(u == v) return u;
    for(int k = LOG - 1; k >= 0; --k)
        if(par[u][k] != par[v][k]) u = par[u][k], v = par[v][k];
    return par[u][0];
}

```

3.4 Sack

```

const int N = 100100;

vector<int> G[N];

int timer = 0;
int tin[N], tout[N], ord[N], sz[N];

void dfs_tour(int u, int p = -1) {
    tin[u] = ++timer;
    ord[timer] = u;
    sz[u] = 1;
    for(const int& v : G[u])
        if(v != p) {
            dfs_tour(v, u);
            sz[u] += sz[v];
        }
    tout[u] = timer;
}

void DFS(int u, int p = -1, bool keep = false) {
    int heavy = -1;
    for(const int& v : G[u])
        if(v != p) {
            if(heavy == -1 || sz[heavy] < sz[v])

```

```

        heavy = v;
    }

    for(const int& v : G[u]) if(v != p && v != heavy) DFS(v, u, 0);

    if(~heavy) DFS(heavy, u, 1);

    for(const int& v : G[u]) if(v != p && v != heavy) {
        for(int i = tin[v]; i <= tout[v]; ++i) {
            // Insert(ord[i]);
        }
    }

    // Insert(u);

    if(!keep) {
        for(int i = tin[u]; i <= tout[u]; ++i) {
            // Remove(ord[i]);
        }
    }
}

```

3.5 Max Matching (Hopcroft)

```

/*
hopcroft karp for finding maximum matching on bipartite graphs
time complexity : O(E.sqrt(V))
layL[i] is the bfs layer of the ith vertex of left partition
layR[i] is for the ith vertex of the right partition
mtR[i] is the vertex matched with the ith vertex of right
partition, -1 if unmatched
adj[i] is list of neighbours of ith vertex of left partition
*/

struct hopcroft {
    ll nl, nr;

    // adj list of the left partition
    vector<vector<int>> adj;
    vector<int> layL, layR, mtR, cur, nxt;

    vector<bool> vis[2], mark;

    hopcroft(int n, int m) : nl(n), nr(m) {
        adj.assign(n, {});
    }

    bool dfs(int u, int len) {
        if(layL[u] != len) return 0;

        layL[u] = -1;
        for (int v : adj[u]) {
            if (layR[v] == len + 1) {
                layR[v] = 0;
                if (mtR[v] == -1 || dfs(mtR[v], len + 1))
                    return mtR[v] = u, 1;
            }
        }

        return 0;
    }

    ll max_matching() {
        layL.assign(nl, 0);
        layR.assign(nr, 0);
        mtR.assign(nr, -1);

        ll res = 0;

        while (true) {
            fill(all(layL), 0);
            fill(all(layR), 0);
            cur.clear();

            for (int u : mtR)
                if(u != -1) layL[u] = -1;

            for (int i = 0; i < sz(adj); i++)

```

```

        if (layL[i] != -1)
            cur.pb(i);

    bool isLast = false;
    for (int lay = 1; ; lay++) {
        nxt.clear();

        for (int u : cur) {
            for (int v : adj[u]) {
                if (mtR[v] == -1) {
                    layR[v] = lay;
                    isLast = true;
                }
                else if (mtR[v] != u && !layR[v]) {
                    layR[v] = lay;
                    nxt.pb(mtR[v]);
                }
            }
        }

        if (isLast) break;

        if (nxt.empty()) return res;

        for (int u : nxt)
            layL[u] = lay;

        swap(cur, nxt);

        for (int i = 0; i < sz(adj); i++)
            res += dfs(i, 0);
    }

    void dfs2(int u, int l) {
        vis[l][u] = true;

        if (!l) {
            for (int v : adj[u]) {
                if (!vis[1][v])
                    dfs2(v, 1);
            }
        }
        else {
            if (mtR[u] != -1 && !vis[0][mtR[u]])
                dfs2(mtR[u], 0);
        }
    }

    //edges in matching -> right to left, else left to right
    //return {left/right, index} of minimum cover
    vector<pll> minCover() {
        vis[0].assign(nl, false);
        vis[1].assign(nr, false);
        mark.assign(nl, false);

        vector<pll> res;
        for (int i = 0; i < nr; i++)
            if (mtR[i] != -1)
                mark[mtR[i]] = true;

        for (int i = 0; i < nl; i++)
            if (!mark[i])
                dfs2(i, 0);

        //unvisited of the left and visited of the right is in
        min cover
        for (int i = 0; i < nl; i++)
            if (!vis[0][i])
                res.pb({0, i});

        for (int i = 0; i < nr; i++)
            if (vis[1][i])
                res.pb({1, i});

        return res;
    }
};

```

3.6 Max Matching (Blossom)

```

/* Complexity: O(E*sqrt(V))
*/
struct Blossom {
    static const int MAXV = 1e3 + 5;
    static const int MAXE = 1e6 + 5;
    int n, E, lst[MAXV], next[MAXE], adj[MAXE];
    int nxt[MAXV], mat[MAXV], dad[MAXV], col[MAXV];
    int que[MAXV], qh, qt;
    int vis[MAXV], act[MAXV];
    int tag, total;

    void init(int n) {
        this->n = n;
        for (int i = 0; i <= n; i++) {
            lst[i] = nxt[i] = mat[i] = vis[i] = 0;
        }
        E = 1, tag = total = 0;
    }

    void add(int u, int v) {
        if (!mat[u] && !mat[v]) mat[u] = v, mat[v] = u,
            total++;
        E++, adj[E] = v, next[E] = lst[u], lst[u] = E;
        E++, adj[E] = u, next[E] = lst[v], lst[v] = E;
    }

    int lca(int u, int v) {
        tag++;
        for(;; swap(u, v)) {
            if (u) {
                if (vis[u = dad[u]] == tag) {
                    return u;
                }
                vis[u] = tag;
                u = nxt[mat[u]];
            }
        }
    }

    void blossom(int u, int v, int g) {
        while (dad[u] != g) {
            nxt[u] = v;
            if (col[mat[u]] == 2) {
                col[mat[u]] = 1;
                que[++qt] = mat[u];
            }
            if (u == dad[u]) dad[u] = g;
            if (mat[u] == dad[mat[u]]) dad[mat[u]] = g;
            v = mat[u];
            u = nxt[v];
        }
    }

    int augment(int s) {
        for (int i = 1; i <= n; i++) {
            col[i] = 0;
            dad[i] = i;
        }
        qh = 0; que[qt = 1] = s; col[s] = 1;
        for (int u, v, i; qh < qt; ) {
            act[u = que[++qh]] = 1;
            for (i = lst[u]; i ; i = next[i]) {
                v = adj[i];
                if (col[v] == 0) {
                    nxt[v] = u;
                    col[v] = 2;
                    if (!mat[v]) {
                        for (; v = u) {
                            u = mat[nxt[v]];
                            mat[v] = nxt[v];
                            mat[nxt[v]] = v;
                        }
                        return 1;
                    }
                }
                col[mat[v]] = 1;
                que[++qt] = mat[v];
            }
        }
        else if (dad[u] != dad[v] && col[v] == 1) {
            int g = lca(u, v);
            blossom(u, v, g);
            blossom(v, u, g);
            for (int j = 1; j <= n; j++) {
                dad[j] = dad[dad[j]];
            }
        }
    }
};

```

```

    }
    }
    }
    return 0;
}
int maxmat() {
    for (int i = 1; i <= n; i++) {
        if (!mat[i]) {
            total += augment(i);
        }
    }
    return total;
}
}

```

3.7 Max Flow (Push Relabel)

```

/**
 * Author: Simon Lindholm
 * Date: 2015-02-24
 * License: CC0
 * Source: Wikipedia, tinyKACTL
 * Description: Push-relabel using the highest label selection
 * rule and the gap heuristic. Quite fast in practice.
 * To obtain the actual flow, look at positive values only.
 * Time:  $\mathcal{O}(V^2\sqrt{E})$ 
 * Status: Tested on Kattis and SPOJ, and stress-tested
 */
#pragma once

struct PushRelabel {
    struct Edge {
        int dest, back;
        ll f, c;
    };
    vector<vector<Edge>> g;
    vector<ll> ec;
    vector<Edge*> cur;
    vector<vi> hs; vi H;
    PushRelabel(int n) : g(n), ec(n), cur(n), hs(2*n), H(n) {}

    void addEdge(int s, int t, ll cap, ll rcap=0) {
        if (s == t) return;
        g[s].push_back({t, sz(g[t]), 0, cap});
        g[t].push_back({s, sz(g[s])-1, 0, rcap});
    }

    void addFlow(Edge& e, ll f) {
        Edge &back = g[e.dest][e.back];
        if (!ec[e.dest] && f) hs[H[e.dest]].push_back(e.dest);
        e.f += f; e.c -= f; ec[e.dest] += f;
        back.f -= f; back.c += f; ec[back.dest] -= f;
    }

    ll calc(int s, int t) {
        int v = sz(g); H[s] = v; ec[t] = 1;
        vi co(2*v); co[0] = v-1;
        rep(i,0,v) cur[i] = g[i].data();
        for (Edge& e : g[s]) addFlow(e, e.c);

        for (int hi = 0;;) {
            while (hs[hi].empty()) if (!hi--) return -ec[s];
            int u = hs[hi].back(); hs[hi].pop_back();
            while (ec[u] > 0) // discharge u
                if (cur[u] == g[u].data() + sz(g[u])) {
                    H[u] = 1e9;
                    for (Edge& e : g[u]) if (e.c && H[u] > H[e.dest]+1)
                        H[u] = H[e.dest]+1, cur[u] = &e;
                    if (++co[H[u]], !--co[hi] && hi < v)
                        rep(i,0,v) if (hi < H[i] && H[i] < v)
                            --co[H[i]], H[i] = v + 1;
                    hi = H[u];
                } else if (cur[u]->c && H[u] == H[cur[u]->dest]+1)
                    addFlow(*cur[u], min(ec[u], cur[u]->c));
                else ++cur[u];
        }
    }

    bool leftOfMinCut(int a) { return H[a] >= sz(g); }
};

```

3.8 Gomory Hu

```

const int INF = 1000000000;

struct Edge {
    int a, b, cap, flow;
};

struct MaxFlow {
    int n, s, t;
    vector<int> d, ptr, q;
    vector< Edge > e;
    vector< vector<int> > g;
    MaxFlow() = default;
    MaxFlow(int _n) : n(_n), d(_n), ptr(_n), q(_n), g(_n) {
        e.clear();
        for (int i = 0; i < n; i++) {
            g[i].clear();
            ptr[i] = 0;
        }
    }

    void addEdge(int a, int b, int cap) {
        Edge e1 = { a, b, cap, 0 };
        Edge e2 = { b, a, 0, 0 };
        g[a].push_back( (int) e.size() );
        e.push_back(e1);
        g[b].push_back( (int) e.size() );
        e.push_back(e2);
    }

    int getMaxFlow(int _s, int _t) {
        s = _s; t = _t;
        int flow = 0;
        for (;;) {
            if (!bfs()) break;
            std::fill(ptr.begin(), ptr.end(), 0);
            while (int pushed = dfs(s, INF))
                flow += pushed;
        }
        return flow;
    }

private:
    bool bfs() {
        int qh = 0, qt = 0;
        q[qt++] = s;
        std::fill(d.begin(), d.end(), -1);
        d[s] = 0;
        while (qh < qt && d[t] == -1) {
            int v = q[qh++];
            for (int i = 0; i < (int) g[v].size(); i++) {
                int id = g[v][i], to = e[id].b;
                if (d[to] == -1 && e[id].flow < e[id].cap) {
                    q[qt++] = to;
                    d[to] = d[v] + 1;
                }
            }
        }
        return d[t] != -1;
    }

    int dfs (int v, int flow) {
        if (!flow) return 0;
        if (v == t) return flow;
        for (; ptr[v] < (int)g[v].size(); ++ptr[v]) {
            int id = g[v][ptr[v]],
                to = e[id].b;
            if (d[to] != d[v] + 1) continue;
            int pushed = dfs(to, min(flow, e[id].cap -
                e[id].flow));
            if (pushed) {
                e[id].flow += pushed;
                e[id-1].flow -= pushed;
                return pushed;
            }
        }
        return 0;
    }
};

const int N = 202;
int ok[N], cap[N][N];
int answer[N][N], parent[N];
int n;
MaxFlow flow;

```

```

void Init(int vertices) {
    n = vertices;
    flow = MaxFlow(vertices);
    for(int i = 0; i < vertices; ++i) ok[i] = parent[i] = 0;
    for(int i = 0; i < vertices; ++i)
        for(int j = 0; j < vertices; ++j)
            cap[i][j] = 0, answer[i][j] = INF;
}

void bfs(int start) {
    memset(ok, 0, sizeof ok);
    queue<int> qu;
    qu.push(start);
    while (!qu.empty()) {
        int u = qu.front(); qu.pop();
        for(int xid = 0; xid < (int) flow.g[u].size(); ++xid) {
            int id = flow.g[u][xid];
            int v = flow.e[id].b, F = flow.e[id].flow, C = flow.e[id].cap;
            if (!ok[v] && F < C) {
                ok[v] = 1;
                qu.push(v);
            }
        }
    }
}

void FindMaxFlow() {
    for(int i = 1; i <= n-1; ++i) {
        flow = MaxFlow(n);
        for(int u = 0; u < n; ++u)
            for(int v = 0; v < n; ++v)
                if (cap[u][v])
                    flow.addEdge(u, v, cap[u][v]);

        int f = flow.getMaxFlow(i, parent[i]);

        bfs(i);
        for(int j = i+1; j < n; ++j)
            if (ok[j] && parent[j] == parent[i])
                parent[j] = i;

        answer[i][parent[i]] = answer[parent[i]][i] = f;
        for(int j = 0; j < i; ++j)

            answer[i][j] = answer[j][i] = min(f, answer[parent[i]][j]);
    }
}

```

3.9 Min Cost Max Flow

```

int n, m, k, source, sink;
struct FlowEdge {
    int to, rev, id, flow, cap, cost;
};
vector<FlowEdge> adj[MAX_N];
int dist[MAX_N];
bool inQueue[MAX_N];
pii trc[MAX_N];
queue<int> q;
int ans;

void addEdge(int u, int v, int cost, int cap) {
    int szU = adj[u].size();
    int szV = adj[v].size();
    adj[u].pb({v, szV, szU, 0, cap, cost});
    adj[v].pb({u, szU, szV, 0, cap, cost});
}

bool BellmanFord() {
    for (int i = 1; i <= n; i++) {
        dist[i] = inf;
    }
    dist[source] = 0;
    q.push(source);
    inQueue[source] = true;
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        inQueue[u] = false;
    }
}

```

```

        for (auto e : adj[u]) {
            int v = e.to;
            int c = (e.flow >= 0 ? 1 : -1) * e.cost;
            if (e.flow < e.cap && dist[u] + c < dist[v]) {
                dist[v] = dist[u] + c;
                trc[v] = {u, e.id};
                if (!inQueue[v]) {
                    q.push(v);
                }
            }
        }
    }
    return dist[sink] < inf;
}

void inc() {
    int incFlow = inf;

    for (int i = sink; i != source; i = trc[i].fi) {
        int u = trc[i].fi;
        int id = trc[i].se;
        minimize(incFlow, (adj[u][id].flow >= 0 ? adj[u][id].cap - adj[u][id].flow : -adj[u][id].flow));
    }

    minimize(incFlow, k);

    for (int i = sink; i != source; i = trc[i].fi) {
        int u = trc[i].fi;
        int id = trc[i].se;
        adj[u][id].flow += incFlow;
        adj[i][adj[u][id].rev].flow -= incFlow;
    }

    ans += incFlow * dist[sink];
    k -= incFlow;

    if (!k) {
        cout << ans << "\n";
        for (int i = 1; i <= n; i++) {
            for (auto e : adj[i]) {
                if (e.flow > 0) {
                    cout << i << " " << e.to << " " << e.flow << "\n";
                }
            }
        }
        cout << "0 0 0";
        exit(0);
    }
}

signed main() {
    ios_base::sync_with_stdio(false);
    cin.tie(nullptr);

    cin >> n >> m >> k >> source >> sink;
    for (int i = 1; i <= m; i++) {
        int u, v, c, d;
        cin >> u >> v >> c >> d;
        addEdge(u, v, c, d);
    }

    while (BellmanFord()) {
        inc();
    }

    cout << -1;

    return 0;
}

// How to find a solution of a Flow Network with Lower Bound
// of flows:
// Add vertices s and t
// Add edge ts with INF capacity
// If there is edge xy with capacity w and lower bound l:
// Add edge xy with capacity w - l
// Let bal[x] be the flow that must enter x inferred from the
// lower bounds
// If bal[x] > 0 add edge sx with capacity bal[x]

```

```
// If bal[x] < 0 add edge xy with capacity -bal[x]
// Notice this finds ANY flow, not necessarily a max flow
```

3.10 Weighted Matching (Hungarian)

```
/**
 * Author: Benjamin Qi, chilli
 * Date: 2020-04-04
 * License: CCO
 * Description: Given a weighted bipartite graph, matches
every node on
 * the left with a node on the right such that no
 * nodes are in two matchings and the sum of the edge weights
is minimal. Takes
 * cost[N][M], where cost[i][j] = cost for L[i] to be matched
with R[j] and
 * returns (min cost, match), where L[i] is matched with
 * R[match[i]]. Negate costs for max cost. Requires N <= M.
 * Time: O(N^2M)
 * Status: Tested on kattis:cordonbleu, stress-tested
 */
#pragma once
```

```
pair<int, vi> hungarian(const vector<vi> &a) {
    if (a.empty()) return {0, {}};
    int n = sz(a) + 1, m = sz(a[0]) + 1;
    vi u(n), v(m), p(m), ans(n - 1);
    rep(i, 1, n) {
        p[0] = i;
        int j0 = 0; // add "dummy" worker 0
        vi dist(m, INT_MAX), pre(m, -1);
        vector<bool> done(m + 1);
        do { // dijkstra
            done[j0] = true;
            int i0 = p[j0], j1, delta = INT_MAX;
            rep(j, 1, m) if (!done[j]) {
                auto cur = a[i0 - 1][j - 1] - u[i0] - v[j];
                if (cur < dist[j]) dist[j] = cur, pre[j] = j0;
                if (dist[j] < delta) delta = dist[j], j1 = j;
            }
            rep(j, 0, m) {
                if (done[j]) u[p[j]] += delta, v[j] -= delta;
                else dist[j] -= delta;
            }
            j0 = j1;
        } while (p[j0]);
        while (j0) { // update alternating path
            int j1 = pre[j0];
            p[j0] = p[j1], j0 = j1;
        }
        rep(j, 1, m) if (p[j]) ans[p[j] - 1] = j - 1;
        return {-v[0], ans}; // min cost
    }
}
```

3.11 Max Clique

```
template <size_t max_n>
class Clique
{
    using bits = bitset<max_n>;
    bits MASK, ZERO, ans;
    const bits *e;
    int N;
    // int64_t calls;
    void bk_init()
    {
        ans = ZERO;
        MASK = ZERO;
        MASK.flip();
        // calls = 0;
    }
    void bk(bits use, bits can_start, bits can_other)
    {
        // ++calls;
        if (can_start.none() && can_other.none())
        {
            if (use.count() > ans.count())
                ans = use;
            return;
        }
    }
}
```

```

    }
    bits r = can_start;
    bool fi = 1;
    for (int i = 0; i < N; ++i)
    {
        if (r[i])
        {
            if (fi)
            {
                fi = 0;
                r &= e[i] ^ MASK;
            }
            use[i] = 1;
            bk(use, can_start & e[i], can_other & e[i]);
            use[i] = 0;
            can_start[i] = 0;
            can_other[i] = 1;
        }
    }
}

static Clique &get()
{
    static Clique c;
    return c;
}

public:
    static bits find_clique(bits const *g, const int &n)
    {
        Clique &c = get();
        c.e = g;
        c.N = n;
        c.bk_init();
        bits me;
        c.bk(me, c.MASK, c.ZERO);
        // cerr << "Calls: " << c.calls << "\n";
        // c.calls = 0;
        return c.ans;
    }
    static bits find_clique(vector<bits> const &g)
    {
        return find_clique(g.data(), g.size());
    }
    static bits find_clique(array<bits, max_n> const &g, const
int &n)
    {
        return find_clique(g.data(), n);
    }
};
```

3.12 2 SAT

//source: <https://wiki.vnoi.info/vi/algo/graph-theory/2-SAT>

```
#include <bits/stdc++.h>

using namespace std;

const int maxN = 500500;

int n, m;

vector<int> G[maxN << 1];

int NOT(int x) {
    return x + (x <= n ? n : -n); // -x
}

void add_clause(int u, int v) {
    G[NOT(u)].push_back(v); // -u -> v
    G[NOT(v)].push_back(u); // -v -> u
}

int id[maxN << 1];
int num[maxN << 1], low[maxN << 1];
int timeDFS = 0, scc = 0;
int st[maxN << 1];

void dfs(int u) {
    num[u] = low[u] = ++timeDFS;
    st[++st[0]] = u;
```

```

for(const int& v : G[u]) {
    if(id[v] != 0) continue;
    if(num[v] == 0) {
        dfs(v);
        low[u] = min(low[u], low[v]);
    } else low[u] = min(low[u], num[v]);
}

if(num[u] == low[u]) {
    for(++scc; true; ) {
        int v = st[st[0]--];
        id[v] = scc;
        if(v == u) break;
    }
}

int main() {
    cin.tie(0)->sync_with_stdio(0);

    cin >> n >> m;
    for(int i = 1; i <= m; ++i) {
        int u, v; cin >> u >> v;
        add_clause(u, v);
    }

    for(int i = 1; i <= 2 * n; ++i) {
        if(!id[i]) dfs(i);
    }

    bool answer = 1;
    for(int i = 1; i <= n; ++i) {
        if(id[i] == id[NOT(i)]) answer = 0;
    }
    if(!answer) {
        cout << "IMPOSSIBLE";
        return 0;
    }
    for(int i = 1; i <= n; ++i) cout << (id[i] < id[NOT(i)])
    << " ";
    return 0;
}

```

4 DP

4.1 Divide And Conquer Optimization

```

void compute(int l, int r, int optl, int opttr) {
    if (l > r)
        return;

    int mid = (l + r) >> 1;
    pair<long long, int> best = {LLONG_MAX, -1};

    for (int k = optl; k <= min(mid, opttr); k++) {
        best = min(best, {(k ? dp_before[k - 1] : 0) + C(k,
        mid), k});
    }

    dp_cur[mid] = best.first;
    int opt = best.second;

    compute(l, mid - 1, optl, opt);
    compute(mid + 1, r, opt, opttr);
}

```

4.2 SoS $O(3^N)$

```

// iterate over all the masks
for (int mask = 0; mask < (1<<n); mask++){
    F[mask] = A[0];
    // iterate over all the subsets of the mask
    for(int i = mask; i > 0; i = (i-1) & mask){
        F[mask] += A[i];
    }
}

```

4.3 SoS $O(2^N)$

```

for(int i = 0; i < (1<<N); ++i)
    F[i] = A[i];

for(int i = 0; i < N; ++i)
    for(int mask = 0; mask < (1<<N); ++mask){
        if(mask & (1<<i))
            F[mask] += F[mask^(1<<i)];
    }

```

5 Strings

5.1 KMP

```

vector<int> prefix_function(string s) {
    int n = (int)s.length();
    vector<int> pi(n);
    for (int i = 1; i < n; ++i) {
        int j = pi[i-1];
        while (j > 0 && s[i] != s[j])
            j = pi[j-1];
        if (s[i] == s[j])
            j++;
        pi[i] = j;
    }
    return pi;
}

```

5.2 Z Function

```

vector<int> zfunction(const string& s) {
    int n = (int) s.size();
    vector<int> z(n);
    for(int i = 1, l = 0, r = 0; i < n; ++i) {
        if(i <= r) z[i] = min(r - i, z[i - l]);
        while(i + z[i] < n && s[z[i]] == s[i + z[i]]) ++z[i];
        if(i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
    }
    return z;
}

```

5.3 Aho Corasick (static)

```

const int CAP = 1003, ALPHABET = 26;
int cntTrie = 1;
int fail[CAP], to[CAP][ALPHABET];
bool ending[CAP];

void add_string(const string& s) {
    int u = 1;
    for(const char& c : s) {
        int x = c - 'a';
        if(!to[u][x]) {
            to[u][x] = ++cntTrie;
        }
        u = to[u][x];
    }
    ending[u] = true;
}

void aho_corasick() {
    queue<int> q; q.push(1);
    while(q.size()) {
        int u = q.front(); q.pop();
        for(int x = 0; x < ALPHABET; ++x) {
            int& v = to[u][x];
            if(!v) {
                v = u == 1 ? 1 : to[fail[u]][x];
            } else {
                if(!fail[v]) fail[v] = fail[u];
                fail[v] = u == 1 ? 1 : to[fail[v]][x];
                ending[v] |= ending[fail[v]];
                q.push(v);
            }
        }
    }
}

```

5.4 Aho Corasick (vector)

```

struct TrieNode {
    int pi = 0;
    int child[26] = {0};
};
vector<TrieNode> trie;
vector<vector<int>> adj;

int TrieInsert(const string& s) {
    int p = 0;
    for (int i = 0; i < s.size(); i++) {
        if (!trie[p].child[s[i] - 'a']) {
            trie[p].child[s[i] - 'a'] = trie.size();
            trie.pb(TrieNode());
        }
        p = trie[p].child[s[i] - 'a'];
    }
    return p;
}

void AhoCorasickBuild() {
    queue<int> q;
    for (int i = 0; i < 26; i++) {
        if (trie[0].child[i]) {
            q.push(trie[0].child[i]);
        }
    }
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        for (int i = 0; i < 26; i++) {
            if (!trie[u].child[i]) continue;
            int j = trie[u].pi;
            while (!trie[j].child[i]) {
                if (!j) break;
                j = trie[j].pi;
            }
            trie[trie[u].child[i]].pi = trie[j].child[i];
            q.push(trie[u].child[i]);
        }
        adj[trie[u].pi].pb(u);
    }
}

signed main() {
    trie.pb(TrieNode());

    adj.resize(trie.size());
    AhoCorasickBuild();
}

```

6 Math

6.1 Chinese Remainder Theorem

```

struct gcd_t { ll x, y, d; };

gcd_t e_gcd(ll a, ll b) {
    if (b == 0) return {1, 0, a};

    gcd_t res = e_gcd(b, a % b);

    return {res.y, res.x - res.y * (a / b), res.d};
}

pll crt(vector<ll> r, vector<ll> m) {
    //find x such that for (1 <= i <= n): x = r[i] (mod m[i])
    //return {y, z} where x = y (mod z), z = lcm of vector m
    //all solutions are congruent modulo z

    ll y = r[0], z = m[0];
    for (int i = 1; i < sz(r); i++) {
        gcd_t cur = e_gcd(z, m[i]);

        ll x = cur.x, d = cur.d;

        if ((r[i] - y) % d != 0) return {-1, -1};
    }
}

```

```

//ka = kb (mod kc) => a = b (mod c) if (gcd(k, c) = 1)
//add (x * (r[i] - y) / d * z) to result (with moduli lcm(z, m[i]))
ll tmp = (x * (r[i] - y) / d) % (m[i] / d);
y = y + tmp * z;
z = z / d * m[i];

y %= z;
if (y < 0) y += z;
}

return {y, z};
}

ll inverse(ll a, ll m) {
    gcd_t cur = e_gcd(a, m);

    return (cur.x % m + m) % m;
}

```

6.2 Miller Rabin

```

// From
https://github.com/SnapDragon64/ContestLibrary/blob/master/math.h
// which also has specialized versions for 32-bit and 42-bit
//
// Tested:
// - https://oj.vnoi.info/problem/icpc22_national_c (fastest solution)
// - https://www.spoj.com/problems/P0N/

// Rabin miller {{{
inline uint64_t mod_mult64(uint64_t a, uint64_t b, uint64_t m)
{
    return __int128_t(a) * b % m;
}

uint64_t mod_pow64(uint64_t a, uint64_t b, uint64_t m) {
    uint64_t ret = (m > 1);
    for (;) {
        if (b & 1) ret = mod_mult64(ret, a, m);
        if (!(b >= 1)) return ret;
        a = mod_mult64(a, a, m);
    }
}

// Works for all primes p < 2^64
bool is_prime(uint64_t n) {
    if (n <= 3) return (n >= 2);
    static const uint64_t small[] = {
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43,
        47, 53, 59, 61, 67,
        71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127,
        131, 137, 139,
        149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197,
        199,
    };
    for (size_t i = 0; i < sizeof(small) / sizeof(uint64_t); ++i) {
        if (n % small[i] == 0) return n == small[i];
    }

    // Makes use of the known bounds for Miller-Rabin pseudoprimes.
    static const uint64_t millerrabin[] = {
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37,
    };
    static const uint64_t A014233[] = { // From OEIS.
        2047LL, 1373653LL, 25326001LL, 3215031751LL,
        2152302898747LL,
        3474749660383LL, 341550071728321LL, 341550071728321LL,
        3825123056546413051LL, 3825123056546413051LL,
        3825123056546413051LL, 0,
    };
    uint64_t s = n-1, r = 0;
    while (s % 2 == 0) {
        s /= 2;
        r++;
    }
    for (size_t i = 0, j; i < sizeof(millerrabin) / sizeof(uint64_t); i++) {

```

```

uint64_t md = mod_pow64(millerrabin[i], s, n);
if (md != 1) {
    for (j = 1; j < r; j++) {
        if (md == n-1) break;
        md = mod_mult64(md, md, n);
    }
    if (md != n-1) return false;
}
if (n < A014233[i]) return true;
}
return true;
}
// }}}

```

6.3 Discrete Logarithm

// Computes x which $a^x = b \bmod n$.

```

long long d_log(long long a, long long b, long long n) {
    long long m = ceil(sqrt(n));
    long long aj = 1;
    map<long long, long long> M;
    for (int i = 0; i < m; ++i) {
        if (!M.count(aj))
            M[aj] = i;
        aj = (aj * a) % n;
    }

    long long coef = mod_pow(a, n - 2, n);
    coef = mod_pow(coef, m, n);
    // coef = a ^ (-m)
    long long gamma = b;
    for (int i = 0; i < m; ++i) {
        if (M.count(gamma)) {
            return i * m + M[gamma];
        } else {
            gamma = (gamma * coef) % n;
        }
    }
    return -1;
}

```

6.4 Fast Fourier Transform

```

typedef complex<double> cmplx;
typedef vector<complex<double>> VC;
const double PI = acos(-1);
struct FFT {
    static void fft(VC &u, int sign) {
        int n = u.size();
        double theta = 2. * PI * sign / n;
        for (int m = n; m >= 2; m >>= 1, theta *= 2.) {
            cmplx w(1, 0), wDelta = polar(1., theta);
            for (int i = 0, mh = m >> 1; i < mh; i++) {
                for (int j = i; j < n; j += m) {
                    int k = j + mh;
                    cmplx temp = u[j] - u[k];
                    u[j] += u[k];
                    u[k] = w * temp;
                }
                w *= wDelta;
            }
        }
        for (int i = 1, j = 0; i < n; i++) {
            for (int k = n >> 1; k > (j ^= k); k >>= 1);
            if (j < i) {
                swap(u[i], u[j]);
            }
        }
    }

    static vector<ll> mul(const vector<int> &a, const
vector<int> &b) {
        int newSz = a.size() + b.size() - 1;
        int fftSz = 1;
        while (fftSz < newSz) fftSz <<= 1;
        VC aa(fftSz, 0.), bb(fftSz, 0.);
        for (int i = 0; i < a.size(); i++) aa[i] = a[i];
        for (int i = 0; i < b.size(); i++) bb[i] = b[i];
        fft(aa, 1), fft(bb, 1);

```

```

        for (int i = 0; i < fftSz; i++) aa[i] *= bb[i];
        fft(aa, -1);
        vector<ll> res(newSz);
        for (int i = 0; i < newSz; i++)
            res[i] = (ll)(aa[i].real() / fftSz + 0.5);
        return res;
    }
};

```

6.5 Berlekamp massey

```

template<typename T> vector<T> berlekampMassey(const vector<T>
&sequence) {
    int n = (int)sequence.size(), len = 0, m = 1;
    vector<T> prevBest(n), coefficients(n);
    T prevDelta = prevBest[0] = coefficients[0] = 1;
    for (int i = 0; i < n; i++, m++) {
        T delta = sequence[i];
        for (int j = 1; j <= len; j++) delta +=
coefficients[j] * sequence[i - j];
        if ((long long)delta == 0) continue;
        vector<T> temp = coefficients;
        T coef = delta / prevDelta;
        for (int j = m; j < n; j++) coefficients[j] -= coef *
prevBest[j - m];
        if ((len << 1) <= i)
            len = i + 1 - len, prevBest = temp, prevDelta =
delta, m = 0;
    }
    coefficients.resize(len + 1);
    coefficients.erase(coefficients.begin());
    for (T &x : coefficients) x = -x;
    return coefficients;
}

```

```

template<typename T> T calcKthTerm(
const vector<T> &coefficients, const vector<T> &sequence,
long long k
) {
    assert(coefficients.size() <= sequence.size());
    int n = (int)coefficients.size();

    auto mul = [&](const vector<T> &a, const vector<T> &b) {
        vector<T> res(a.size() + b.size() - 1u);
        for (int i = 0; i < (int)a.size(); i++)
            for (int j = 0; j < (int)b.size(); j++)
                res[i + j] += a[i] * b[j];
        for (int i = (int)res.size() - 1; i >= n; i--)
            for (int j = n - 1; j >= 0; j--)
                res[i - j - 1] += res[i] * coefficients[j];
        res.resize(min((int)res.size(), n));
        return res;
    };

    vector<T> a = (n == 1 ? vector<T>{coefficients[0]} :
vector<T>{0, 1}), x{1};
    for (; k; k >>= 1) {
        if (k & 1) x = mul(x, a);
        a = mul(a, a);
    }
    x.resize(n);
    T res = 0;
    for (int i = 0; i < n; i++) res += x[i] * sequence[i];
    return res;
}

```

// Usual: cout << calcKthTerm(berlekampMassey(ans), ans, n) << "\n";

6.6 Advanced

```

// Copied from
https://github.com/imeplusplus/icpc-notebook/blob/master/math/advance
/* Line integral = integral(sqrt(1 + (dy/dx)^2)) dx */

/* Multiplicative Inverse over MOD for all 1..N - 1 < MOD in
O(N)
Only works for prime MOD. If all 1..MOD - 1 needed, use N =
MOD */
ll inv[N];

```

```

inv[1] = 1;
for(int i = 2; i < N; ++i)
    inv[i] = MOD - (MOD / i) * inv[MOD % i] % MOD;

/* Catalan
f(n) = sum(f(i) * f(n - i - 1)), i in [0, n - 1] = (2n)! / ((n+1)! * n!) = ...
If you have any function f(n) (there are many) that follows
this sequence (0-indexed):
1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786,
208012, 742900, 2674440
than it's the Catalan function */
ll cat[N];
cat[0] = 1;
for(int i = 1; i + 1 < N; i++) // needs inv[i + 1] till inv[N - 1]
    cat[i] = 2ll * (2ll * i - 1) * inv[i + 1] % MOD * cat[i - 1] % MOD;

/* Floor(n / i), i = [1, n], has <= 2 * sqrt(n) diff values.
Proof: i = [1, sqrt(n)] has sqrt(n) diff values.
For i = [sqrt(n), n] we have that 1 <= n / i <= sqrt(n)
and thus has <= sqrt(n) diff values.
*/
/* l = first number that has floor(N / l) = x
r = last number that has floor(N / r) = x
N / r >= floor(N / l)
r <= N / floor(N / l)*/
for(int l = 1, r; l <= n; l = r + 1){
    r = n / (n / l);
    // floor(n / i) has the same value for l <= i <= r
}

/* Recurrence using matrix
h[i + 2] = a1 * h[i + 1] + a0 * h[i]
[h[i] h[i-1]] = [h[1] h[0]] * [a1 1] ^ (i - 1)
                        [a0 0]      */

/* Fibonacci in O(log(N)) with memoization
f(0) = f(1) = 1
f(2*k) = f(k)^2 + f(k - 1)^2
f(2*k + 1) = f(k)*f(k) + 2*f(k - 1) */

/* Wilson's Theorem Extension
B = b1 * b2 * ... * bm (mod n) = +-1, all bi <= n such that
gcd(bi, n) = 1
if(n <= 4 or n = (odd prime)^k or n = 2 * (odd prime)^k) B =
-1; for any k
else B = 1; */

/* Stirling numbers of the second kind
S(n, k) = Number of ways to split n numbers into k non-empty
sets
S(n, 1) = S(n, n) = 1
S(n, k) = k * S(n - 1, k) + S(n - 1, k - 1)
Sr(n, k) = S(n, k) with at least r numbers in each set
Sr(n, k) = k * Sr(n - 1, k) + (n - 1) * Sr(n - r, k - 1)
(r - 1)
S(n - d + 1, k - d + 1) = S(n, k) where if indexes i, j
belong to the same set, then |i - j| >= d */

/* Burnside's Lemma
|Classes| = 1 / |G| * sum(K ^ C(g)) for each g in G
G = Different permutations possible
C(g) = Number of cycles on the permutation g
K = Number of states for each element

Different ways to paint a necklace with N beads and K colors:
G = {(1, 2, ... N), (2, 3, ... N, 1), ... (N, 1, ... N - 1)}
gi = (i, i + 1, ... i + N), (taking mod N to get it right) i
= 1 ... N
i -> 2i -> 3i ..., Cycles in gi all have size n / gcd(i, n),
so C(gi) = gcd(i, n)
Ans = 1 / N * sum(K ^ gcd(i, n)), i = 1 ... N
(For the brave, you can get to Ans = 1 / N * sum(euler_phi(N
/ d) * K ^ d), d | N) */

/* Mobius Inversion
Sum of gcd(i, j), 1 <= i, j <= N?
sum(k->N) k * sum(i->N) sum(j->N) [gcd(i, j) == k], i = a *
k, j = b * k

```

```

= sum(k->N) k * sum(a->N/k) sum(b->N/k) [gcd(a, b) == 1]
= sum(k->N) k * sum(a->N/k) sum(b->N/k) sum(d->N/k) [d | a] *
[d | b] * mi(d)
= sum(k->N) k * sum(d->N/k) mi(d) * floor(N / kd)^2, 1 = kd,
1 <= N, k | 1, d = 1 / k
= sum(l->N) floor(N / l)^2 * sum(k|l) k * mi(l / k)
If f(n) = sum(x|n)(g(x) * h(x)) with g(x) and h(x)
multiplicative, than f(n) is multiplicative
Hence, g(1) = sum(k|1) k * mi(l / k) is multiplicative
= sum(l->N) floor(N / l)^2 * g(1) */

```

```

/* Frobenius / Chicken McNugget
n, m given, gcd(n, m) = 1, we want to know if it's possible to
create N = a * n + b * m
N, a, b >= 0
The greatest number NOT possible is n * m - n - m
We can NOT create (n - 1) * (m - 1) / 2 numbers */

```

7 Geometry

7.1 Geomtry Point

```

struct Point{
    typedef ll T;
    T x, y;
    Point(T _x = 0, T _y = 0) : x(_x), y(_y) {}

    bool operator < (Point p) const { return tie(x, y) <
tie(p.x, p.y); }
    bool operator > (Point p) const { return tie(x, y) >
tie(p.x, p.y); }
    bool operator == (Point p) const { return tie(x, y) ==
tie(p.x, p.y); }
    bool operator != (Point p) const { return tie(x, y) !=
tie(p.x, p.y); }

    Point operator + (Point p) const { return Point(x + p.x, y
+ p.y); }
    Point operator - (Point p) const { return Point(x - p.x, y
- p.y); }
    T operator * (Point p) const { return x * p.x + y * p.y; }
    T operator ^ (Point p) const { return x * p.y - y * p.x; }

    Point operator * (T d) const { return Point(x * d, y * d);
}
    Point operator / (T d) const { return Point(x / d, y / d);
}

    T len2() const { return x * x + y * y; }
    double len() const { return sqrt((double)len2()); }
    Point perp() { return Point(-y, x); }
    friend ostream& operator << (ostream &os, const Point &p)
{
        return os << "(" << p.x << ", " << p.y << ")"; }
};

ll ccw(const Point &P0, const Point &P1, const Point &P2){
    return (P1 - P0) ^ (P2 - P1);
}

ll sgn(const ll &x) { return (x >= 0 ? (x ? 1 : 0) : -1); }

```

7.2 Convex Hull

```

vector<Point> convexHull(vector<Point> dots) {
    sort(dots.begin(), dots.end());
    vector<Point> A(1, dots[0]);
    const int sz = dots.size();
    for(int c = 0; c < 2; reverse(all(dots)), c++)
        for(int i = 1, t = A.size(); i < sz;
            A.emplace_back(dots[i++]))
            while (A.size() > t and ccw(A[A.size()-2], A.back(),
                dots[i]) < 0)
                A.pop_back();
            A.pop_back(); return A;
}

```

7.3 Minimum Enclosing Circle

```
// Minimum Enclosing Circle
//
// O(n) com alta probabilidade

const double EPS = 1e-12;
mt19937 rng((int)
chrono::steady_clock::now().time_since_epoch().count());

struct pt {
    double x, y;
    pt(double x_ = 0, double y_ = 0) : x(x_), y(y_) {}
    pt operator + (const pt& p) const { return pt(x+p.x, y+p.y); }
    pt operator - (const pt& p) const { return pt(x-p.x, y-p.y); }
    pt operator * (double c) const { return pt(x*c, y*c); }
    pt operator / (double c) const { return pt(x/c, y/c); }
};

double dot(pt p, pt q) { return p.x*q.x+p.y*q.y; }
double cross(pt p, pt q) { return p.x*q.y-p.y*q.x; }
double dist(pt p, pt q) { return sqrt(dot(p-q, p-q)); }

pt center(pt p, pt q, pt r) {
    pt a = p-r, b = q-r;
    pt c = pt(dot(a, p+r)/2, dot(b, q+r)/2);
    return pt(cross(c, pt(a.y, b.y)), cross(pt(a.x, b.x), c)) /
        cross(a, b);
}

struct circle {
    pt cen;
    double r;
    circle(pt cen_, double r_) : cen(cen_), r(r_) {}
    circle(pt a, pt b, pt c) {
        cen = center(a, b, c);
        r = dist(cen, a);
    }
    bool inside(pt p) { return dist(p, cen) < r+EPS; }
};

circle minCirc(vector<pt> v) {
    shuffle(v.begin(), v.end(), rng);
    circle ret = circle(pt(0, 0), 0);
    for (int i = 0; i < v.size(); i++) if (!ret.inside(v[i])) {
        ret = circle(v[i], 0);
        for (int j = 0; j < i; j++) if (!ret.inside(v[j])) {
            ret = circle((v[i]+v[j])/2, dist(v[i], v[j])/2);
            for (int k = 0; k < j; k++) if (!ret.inside(v[k]))
                ret = circle(v[i], v[j], v[k]);
        }
    }
    return ret;
}
```

7.4 Manhattan MST

```
vector<array<ll, 3>> manhattanMST(vector<Point> ps) {
    vector<int> id(sz(ps));
    iota(all(id), 0);
    vector<array<ll, 3>> edges;
    for (int k = 0; k < 4; k++) {
        sort(all(id), [&](int i, int j) { return (ps[i] -
            ps[j]).x < (ps[j] - ps[i]).y; });
        map<int, int> sweep;
        for (int i : id) {
            for (auto it = sweep.lower_bound(-ps[i].y);
                it != sweep.end(); sweep.erase(it++)) {
                int j = it->second;
                Point d = ps[i] - ps[j];
                if (d.y > d.x) break;
                edges.push_back({d.y + d.x, i, j});
            }
            sweep[-ps[i].y] = i;
        }
        for (Point &p : ps)
            if (k & 1)
                p.x = -p.x;
            else
```

```
                swap(p.x, p.y);
    }
    return edges;
}
```

7.5 Some Common Geometry Operations

```
ll ccw(const Point &P0, const Point &P1, const Point &P2){
    return (P1 - P0) ^ (P2 - P1);
}

bool on_segment(Point &p, Point &p0, Point &p1){
    if((p1 - p0) * (p - p1) > 0) return false;
    if((p0 - p1) * (p - p0) > 0) return false;
    return (ccw(p, p0, p1) == 0);
}

db dist_segment(Point &p, Point &p0, Point &p1){
    if((p1 - p0) * (p - p1) >= 0) return (p - p1).len();
    if((p0 - p1) * (p - p0) >= 0) return (p - p0).len();
    return abs((db)((p1 - p0) ^ (p - p0)) / (p1 - p0).len());
}

bool insideConvex(Point p, vector<Point> &poly){
    // clock wise
    int n = sz(poly);
    if(ccw(poly[0], poly[1], p) >= 0) return false;
    if(ccw(poly[n-1], poly[0], p) >= 0) return false;

    ll l = 1, r = n-1;
    while(l < r){
        ll mid = (l+r+1)/2;
        if(ccw(poly[0], p, poly[mid]) >= 0) l = mid;
        else r = mid - 1;
    }
    r = l + 1;

    return (ccw(poly[l], p, poly[r]) > 0);
}

ll wn_poly(Point p, vector<Point> &poly){
    // 1 if inside 0 if outside, INF if on boundary
    // counter clock wise
    const ll on_boundary = INF;
    ll wn = 0;

    int n = sz(poly);
    for(int i = 0; i < n; i++){
        if(p == poly[i]) return on_boundary;

        int j = (i + 1 != n ? i + 1 : 0);

        if(poly[i].y == p.y && poly[j].y == p.y){
            if(min(poly[i].x, poly[j].x) <= p.x
                && p.x <= max(poly[i].x, poly[j].x))
                return on_boundary;
        }
        else{
            bool below = (poly[i].y <= p.y);
            //different sides of horizontal ray
            if (below != (poly[j].y <= p.y)){
                ll orientation = ccw(p, poly[i], poly[j]);

                if (orientation == 0) return on_boundary;
                if (below == (orientation > 0)) wn += (below ?
                    1 : -1);
            }
        }
    }

    return wn;
}

bool line_intersect(pii a, pii b, pii c, pii d) {
    if (!ccw(c, a, b) || !ccw(d, a, b) || !ccw(a, c, d) ||
        !ccw(b, c, d)) {
        if (!ccw(c, a, b) && dot_product(a, c, b) <= 0) {
            return true;
        }
        if (!ccw(d, a, b) && dot_product(a, d, b) <= 0) {
            return true;
        }
    }
}
```

```

}
if (!ccw(a, c, d) && dot_product(c, a, d) <= 0) {
    return true;
}
if (!ccw(b, c, d) && dot_product(c, b, d) <= 0) {
    return true;
}
return false;
}
return (ccw(a, b, c) * ccw(a, b, d) < 0 && ccw(c, d, a) *
        ccw(c, d, b) < 0);

```

8 Miscellaneous

8.1 Subset

```

void subset(int x) {
    for(int ms = x; ms > 0; ms = (ms - 1) & x) {
        // ms is the subset
    }
    // not include empty.
}

```

8.2 Parallel Binary Search

```

// Parallel Binary Search - O(nlog n * cost to update data
// structure + qlog n * cost for binary search condition)
struct Query { int i, ans; /* query related info*/ };
vector<Query> req;

void pbs(vector<Query>& qs, int l /* = min value*/, int r /* =
max value*/) {
    if (qs.empty()) return;

    if (l == r) {
        for (auto& q : qs) req[q.i].ans = l;
        return;
    }

    int mid = (l + r) / 2;
    // mid = (l + r + 1) / 2 if different from simple
    // upper/lower bound

    for (int i = l; i <= mid; i++) {
        // add value to data structure
    }

    vector<Query> vl, vr;
    for (auto& q : qs) {
        if (/* cond */) vl.push_back(q);
        else vr.push_back(q);
    }

    pbs(vr, mid + 1, r);

    for (int i = l; i <= mid; i++) {
        // remove value from data structure
    }

    pbs(vl, l, mid);
}

```

8.3 Hilber Order for Mo's

```

inline ll gilbertOrder(int x, int y, int pow, int rotate) {
    if (pow == 0) {
        return 0;
    }
    int hpow = 1 << (pow - 1);
    int seg = (x < hpow) ? ((y < hpow) ? 0 : 3) : ((y < hpow)
? 1 : 2);
    seg = (seg + rotate) & 3;
    const int rotateDelta[4] = {3, 0, 0, 1};
    int nx = x & (x ^ hpow), ny = y & (y ^ hpow);
    int nrot = (rotate + rotateDelta[seg]) & 3;
    ll subSquareSize = 1ll(1) << (2 * pow - 2);
    ll ans = seg * subSquareSize;
    ll add = gilbertOrder(nx, ny, pow - 1, nrot);
    ans += (seg == 1 || seg == 2) ? add : (subSquareSize - add
- 1);
}

```

```

return ans;
}

struct Query {
    ll l, r, id, ord;
    Query(int _l, int _r, int _id) : l(_l), r(_r), id(_id) {}
    inline void calc() {
        ord = gilbertOrder(l, r, 21, 0);
    }
};

inline bool operator<(const Query &a, const Query &b)
{
    return a.ord < b.ord;
}

// sort(Q + 1, Q + num_q + 1);
// while(l > Q[i].l) adding(--l);
// while(r < Q[i].r) adding(++r);
// while(l < Q[i].l) erase(l++);
// while(r > Q[i].r) erase(r--);

```

8.4 Mo with Update

```

// for(int i = 1; i <= n; ++i)
//     Bucket[i] = i / SIZE_BLOCK;

// SIZE_BLOCK = pow(n, 0.71);
// UPDATE_BLOCK = max(1.00, pow(C2, 0.71));

int P[N], Bucket[N], SIZE_BLOCK = 0, UPDATE_BLOCK = 0;
Query Q[N];
Update Q2[N];

struct Query {
    int data, l, r, t;
    tuple<int, int, int> mo() {
        return {t / UPDATE_BLOCK, ((t / UPDATE_BLOCK) & 1 ? 1
: -1) * Bucket[l], Bucket[l] & 1 ? r : -r};
    }
};

struct Update {
    int i, bef, now;
};

// iota(P + 1, P + num_q + 1, 1);
// sort(P + 1, P + num_q + 1, [&] (int u, int v) { return
Q[u].mo() < Q[v].mo(); });

for(int i = 1; i <= num_q; ++i) {
    int id = P[i];

    while(timer < Q[id].t) {
        ++timer;
        update(Q2[timer].i, Q2[timer].now);
    }

    while(timer > Q[id].t) {
        update(Q2[timer].i, Q2[timer].bef);
        --timer;
    }

    while(l > Q[id].l) adding(ID[--l]);
    while(r < Q[id].r) adding(ID[++r]);
    while(l < Q[id].l) erase(ID[l++]);
    while(r > Q[id].r) erase(ID[r--]);
}

```

8.5 SQRT Decomposition

```

void apply_lazy(int b) {
    for (int i = b * S; i < (b + 1) * S; i++) {
        // ...
    }

    lazy[b] = 0;
}

void manual_update(int l, int r, int x) {
    int b = l / S;
    for (int i = l; i <= r; i++) {

```

```

    /// ...
}
}

void update(int l, int r, int x) {
    int block_l = l / S;
    int block_r = r / S;

    if (block_l == block_r) {
        apply_lazy(block_l);
        manual_update(l, r, x);
    } else {
        update_block(block_l, block_r, x);

        apply_lazy(block_l);
        apply_lazy(block_r);

        manual_update(l, (block_l + 1) * S - 1, x);
        manual_update(block_r * S, r, x);
    }
}

```

8.6 Permutation to Integer

```

/**
 * Author: Simon Lindholm
 * Date: 2018-07-06
 * License: CC0
 * Description: Permutation -> integer conversion. (Not order
preserving.)
 * Integer -> permutation can use a lookup table.
 * Time: O(n)
 */
using vi = vector<int>;
int permToInt(vi& v) {
    int use = 0, i = 0, r = 0;
    for(int x:v) r = r * ++i + __builtin_popcount(use &
-(1<<x)),
        use |= 1 << x; // (note: minus, not
~!)
    return r;
}

```

9 Math Extra

9.1 Combinatorial formulas

$$\begin{aligned}
 \sum_{k=0}^n k^2 &= n(n+1)(2n+1)/6 \\
 \sum_{k=0}^n k^3 &= n^2(n+1)^2/4 \\
 \sum_{k=0}^n k^4 &= (6n^5 + 15n^4 + 10n^3 - n)/30 \\
 \sum_{k=0}^n k^5 &= (2n^6 + 6n^5 + 5n^4 - n^2)/12 \\
 \sum_{k=0}^n x^k &= (x^{n+1} - 1)/(x - 1) \\
 \sum_{k=0}^n kx^k &= (x - (n+1)x^{n+1} + nx^{n+2})/(x - 1)^2 \\
 \binom{n}{k} &= \frac{n!}{(n-k)!k!} \\
 \binom{n}{k} &= \binom{n-1}{k} + \binom{n-1}{k-1} \\
 \binom{n}{k} &= \frac{n}{n-k} \binom{n-1}{k} \\
 \binom{n}{k} &= \frac{n-k+1}{k} \binom{n}{k-1} \\
 \binom{n+1}{k} &= \frac{n+1}{n-k+1} \binom{n}{k} \\
 \binom{n}{k+1} &= \frac{n-k}{k+1} \binom{n}{k} \\
 \sum_{k=1}^n k \binom{n}{k} &= n2^{n-1} \\
 \sum_{k=1}^n k^2 \binom{n}{k} &= (n + n^2)2^{n-2} \\
 \binom{m+n}{r} &= \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} \\
 \binom{n}{k} &= \prod_{i=1}^k \frac{n-k+i}{i}
 \end{aligned}$$

9.2 Number theory identities

Lucas' Theorem: For non-negative integers m and n and a prime p ,

$$\binom{m}{n} \equiv \prod_{i=0}^k \binom{m_i}{n_i} \pmod{p},$$

where

$$m = m_k p^k + m_{k-1} p^{k-1} + \cdots + m_1 p + m_0$$

is the base p representation of m , and similarly for n .

9.3 Stirling Numbers of the second kind

Number of ways to partition a set of n numbers into k non-empty subsets.

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{(k-j)} \binom{k}{j} j^n$$

Recurrence relation:

$$\begin{aligned}
 \left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} &= 1 \\
 \left\{ \begin{matrix} n \\ 0 \end{matrix} \right\} &= \left\{ \begin{matrix} 0 \\ n \end{matrix} \right\} = 1 \\
 \left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} &= k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} + \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}
 \end{aligned}$$

9.4 Burnside's Lemma

Let G be a finite group that acts on a set X . For each g in G let X^g denote the set of elements in X that are fixed by g , which means $X^g = \{x \in X | g(x) = x\}$. Burnside's lemma asserts the following formula for the number of orbits, denoted $|X/G|$:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

9.5 Numerical integration

RK4: to integrate $\dot{y} = f(t, y)$ with $y_0 = y(t_0)$, compute

$$\begin{aligned}
 k_1 &= f(t_n, y_n) \\
 k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_1\right) \\
 k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_2\right) \\
 k_4 &= f(t_n + h, y_n + h k_3) \\
 y_{n+1} &= y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)
 \end{aligned}$$